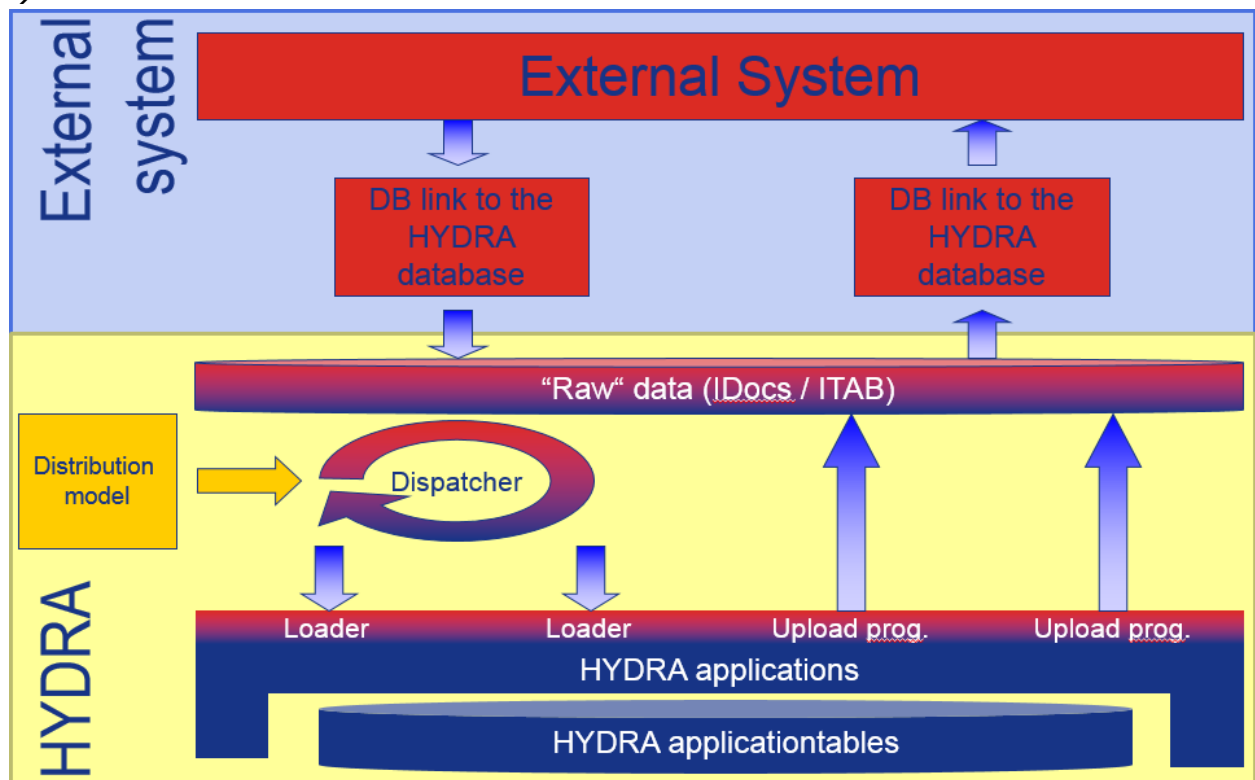


1 Interface based on database - technical instructions

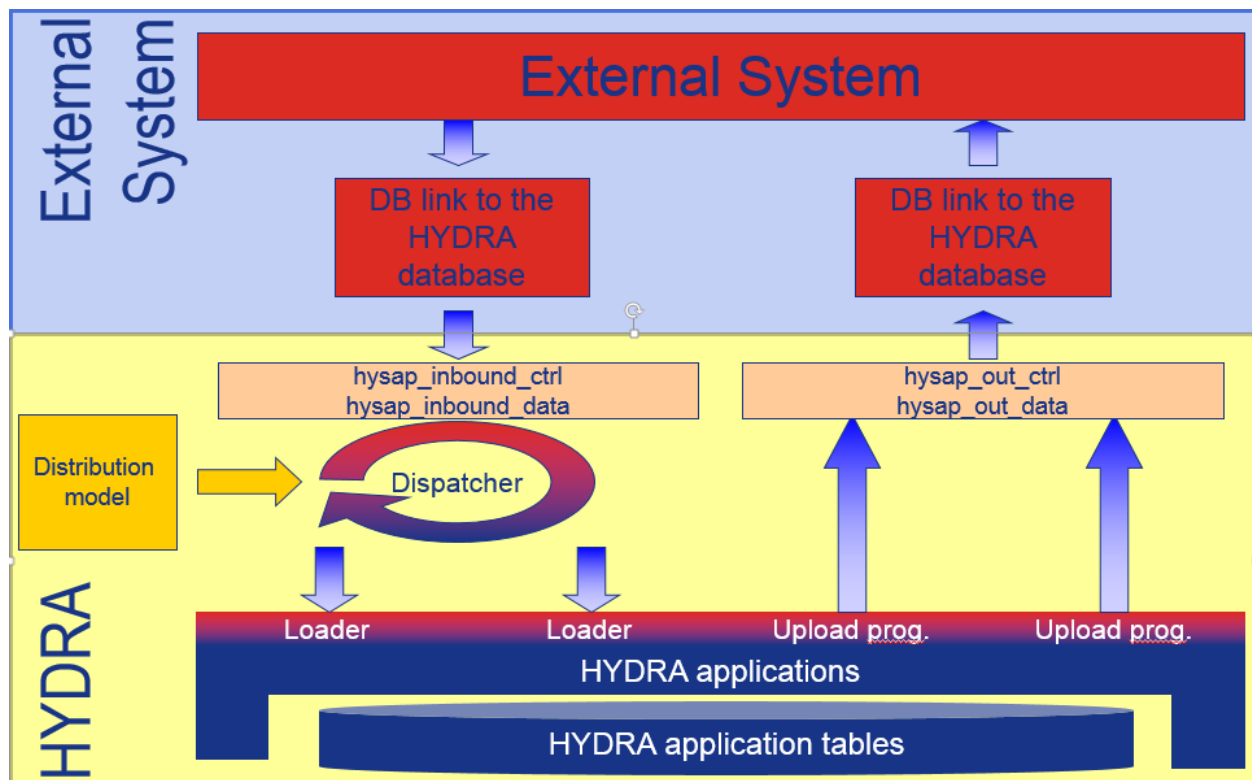
Basic structure of the interface

The database-based interface enables external applications to file and retrieve data for the data exchange with HYDRA from interface tables of the MES Weaver MLE layer. This document describes the database structure and the process of exchanging data.

→



The MLE layer includes four tables to process data transfer.



HYDRA manages (IDoc) data from the ERP system and IDocs to be uploaded/confirmed in the presented interface tables. Data is filed according to the relevant interface structure. When it comes to HYDRA inbound processing, data from these tables is added to the HYDRA data model. With HYDRA outbound processing, upload data is written into outbound tables where external systems can retrieve it.

Data supply external system --> HYDRA

The external application enters data into the corresponding tables of the HYDRA MLE layer. It is important that data (1-n data records) for the IDoc is first written in the table `hysap_inbound_data` by the external application. Then the external application writes a control record (1 data record) including the relevant data in the table `hysap_inbound_ctrl`. The external application links the entries of both tables by a distinct transaction number.

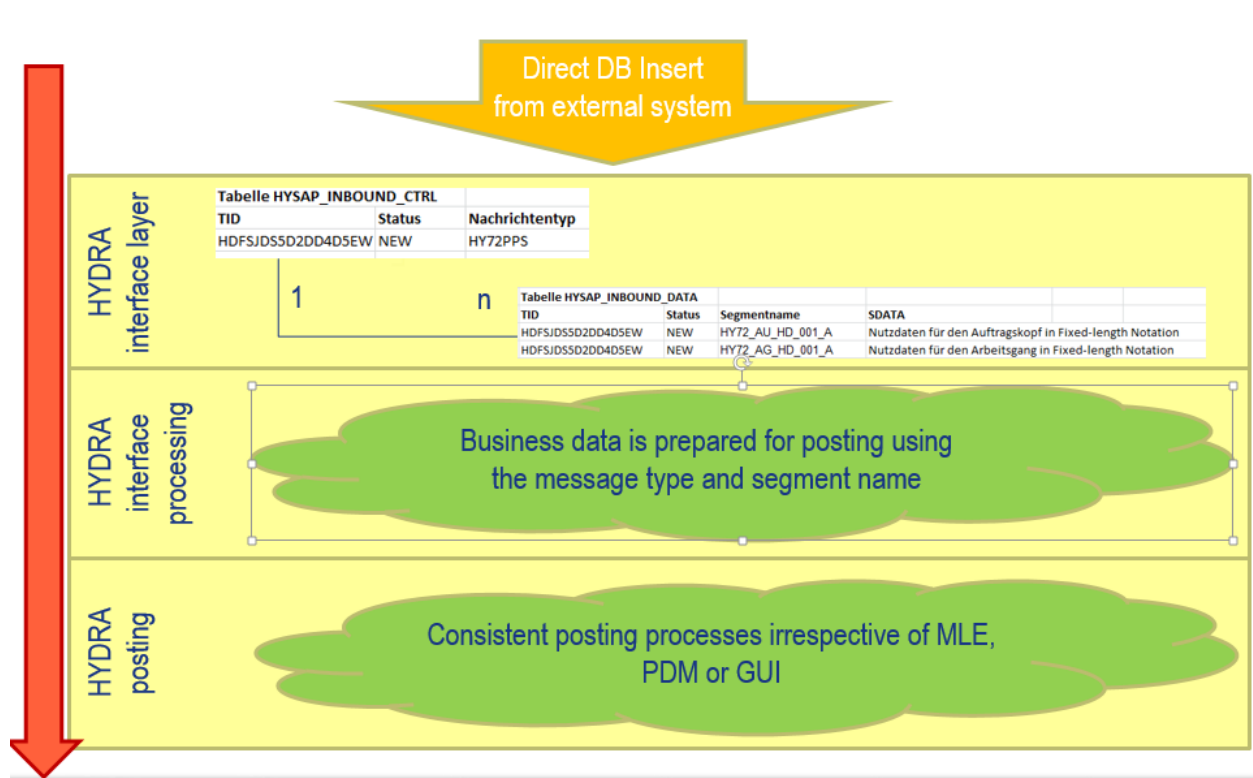
The transaction number must be distinct and structured as described below:



DBLINK<user-defined section>

We recommend using a date/time stamp of format "DBLINKYYYYMMDDHHMMSSsss" for the user-defined section. But different structures are also allowed, as long as the number is distinct within the tables "`hysap_inbound_ctrl`" and "`hysap_inbound_data`".

The HYDRA MLE Dispatcher organizes inbound processing in HYDRA. This dispatcher monitors inbound transactions. When new messages arrive, it also specifies and starts the respective processing routine (program) based on the message type (from the MLE distribution model) to transfer data to HYDRA. Inbound transactions are processed according to the sequence specified by the external application. Consequently, a transaction can only be processed, once the previous transaction has been completed. Log and error files are created for data transferred by HYDRA.



The tasks of the individual steps in the HYDRA inbound processing are divided as follows between the external system and HYDRA:

Step	Responsible system
Write data segments (table hysap_inbound_data)	External system
Write the control record (table hysap_inbound_ctrl)	External system
Processing of data	HYDRA (MLE dispatcher + processing programs)

Data retrieval HYDRA → external system

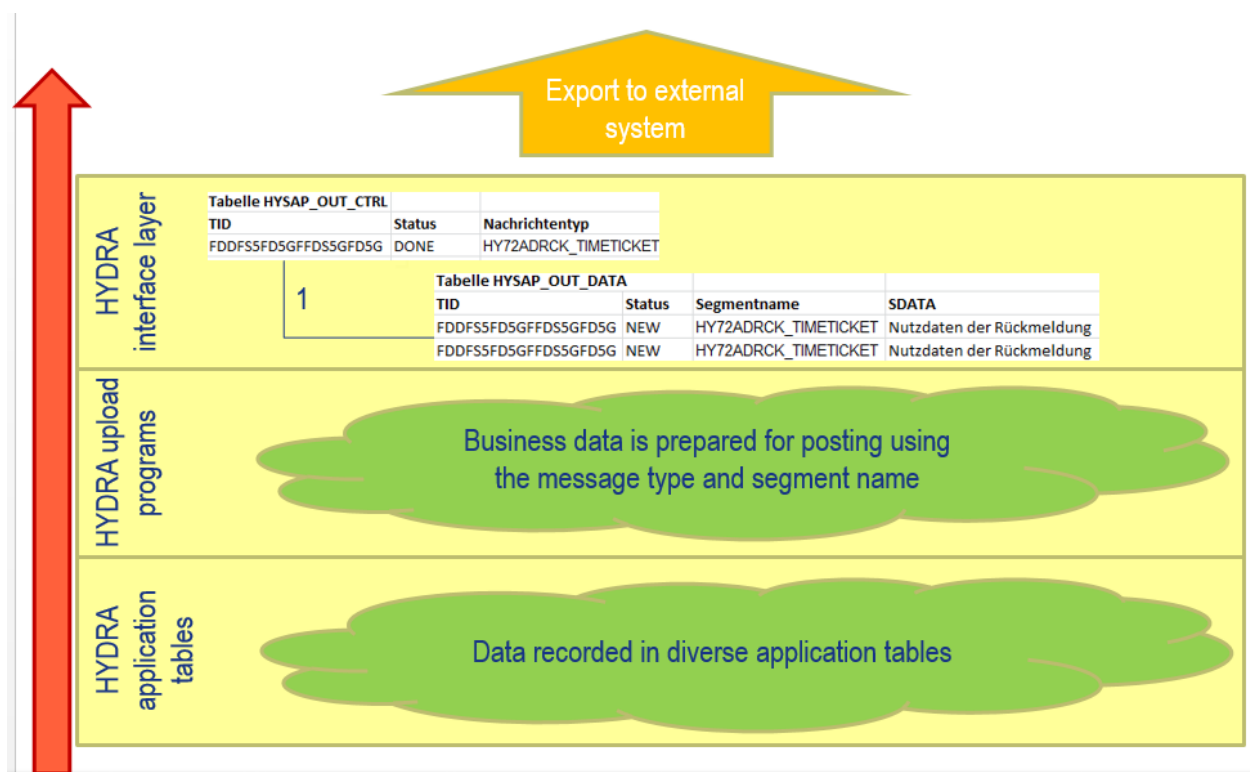
HYDRA's upload programs provide data to be uploaded in the relevant interface format into the outbound table "hysap_out_data". From there the external application can export the table to the other system. After data has been exported, the external application updates specific fields in the outbound table hysap_out_data (see details in the table description of table hysap_out_data). The external application then generates a control record in the table hysap_out_ctrl (see details in the table description of table hysap_out_ctrl) and links the two tables with a distinct transaction number.

The transaction number must be distinct and structured as described below:



DBLINK<user-defined section>

We recommend using a date/time stamp of format "DBLINKYYYYMMDDHHMMSSsss" for the user-defined section. But different structures are also allowed, as long as the number is distinct within the tables "hysap_out_ctrl" and "hysap_out_data".



The uploaded records are prepared by HYDRA's upload programs, converted into the IDoc format and stored in the table "hysap_out_data". Records that have not yet been uploaded have the status "000" (hysap_out_data.ds_status).

During the transfer the external application must change the status (hysap_out_data.DS_STATUS) to "100". Once transferred successfully, the external application must change the status to "099" in the table "hysap_out_data". An entry is to be generated in the table "hysap_out_ctrl" and both tables must be linked with a distinct transaction number.

Processing of multi-level, hierarchical outbound structures represents an exception. The HYDRA upload programs provide these structures in the appropriate format required by the respective interface specification. The external application connects the header data record and the detailed records (sub-segments) via the fields VERWEIS and KOPF_VERWEIS of the table hysap_out_data. In this context, the KOPF_VERWEIS of detailed records includes the header record's reference.

Example:

The following hierarchical structure is defined in the specification of the interface:

Segments:

Z2WEI000X000 (goods receipt)

Z2CNRATT_C000X000 (alphanumeric batch attributes part 1)

Z2CNRATT_C001X000 (alphanumeric batch attributes part 2)

Z2CNRATT_N001X000 (numeric batch attributes)

Z2CNRBAUM000X000 (optionally: tree of generation)

Z2TOLO000X000 (optionally: sub-batches)

Z2CNR_USRFLD000X000 (optionally: user fields)

HYDRA's upload programs provide data in the following format in the table "hysap_out_data".

Table HYSAP_OUT_DATA				
Status	Segment name	SDATA	VERWEIS	KOPF_VERWEIS
000	Z2WEI000X000	User data of upload	52143	
000	Z2CNRATT_C000X000	User data of upload	52144	52143
001	Z2CNRATT_C001X000	User data of upload	52145	52143
001	Z2CNRATT_N000X000	User data of upload	52146	52143
001	Z2WEI000X000	User data of upload	52147	
001	Z2CNRATT_C000X000	User data of upload	52148	52147
001	Z2CNRATT_C001X000	User data of upload	52149	52147
001	Z2CNRATT_N000X000	User data of upload	52150	52147
001	Z2WEI000X000	User data of upload	52151	
001	Z2CNRATT_C000X000	User data of upload	52152	52151
001	Z2CNRATT_C001X000	User data of upload	52153	52151
001	Z2CNRATT_N000X000	User data of upload	52154	52151

In order to export data, external applications must select header records at first and set their status to "100". Now the sub-segments (detailed records) can be selected using the "KOPF_VERWEIS" column. Those lines matching the value of the "VERWEIS" column of the header record (in our example: "52143") will also be selected.

The tasks of the individual steps in the HYDRA outbound processing are divided as follows between the external system and HYDRA:

Step	Responsible system
Write data segments (table hysap_out_data with hysap_out_data.status = ,000')	HYDRA upload programs
Set data records to status „IN PROCESS“ (table hysap_out_data with hysap_out_data.status = ,100')	External system
Update of specific fields in the data records (table hysap_out_data)	External system
Write the control record (table hysap_out_ctrl)	External system
Link of data records and control record via transaction number	External system

HYDRA inbound processing – table hysap_inbound_ctrl

The table "hysap_inbound_ctrl" includes the control records of the data records transferred to HYDRA. The table contains fields of the IDoc control record and additional control fields for HYDRA processing. The structure of the table and the meaning of single fields are described in the sections that follow.

Field name	Data type	Description	Purpose	Mandatory field
ta_id	CHAR(30)	Transaction ID	Transaction number Unique key (see above note)	X
ta_type	CHAR(5)	Description of the type of structure	fixed "IDOC"	X
ta_status	CHAR(3)	Processing status	Fixed "000"	X
ta_logsys	CHAR(10)	Logical system	Not used / assigned	
ta_lines	INTEGER	Number of data records included in the transaction	Number of data records of the IDoc	X
ta_idone	INTEGER	Number of data records processed in the transaction	Fixed "0"	X
ta_lunknown	INTEGER	Number of unknown data records included in the transaction	Fixed "0"	X
ta_lerror	INTEGER	Number of faulty data records included in the transaction	Fixed "0"	X
ta_savdate	DATE	Date of receipt in HYDRA	Current date in format "mm/dd/yyyy"	X
ta_savtime	INTEGER	Time of receipt in HYDRA	Current time in "seconds after midnight"	X
ta_workdate	DATE	Date of processing	Not used / assigned	
ta_worktime	INTEGER	Processing time	Not used / assigned	
sap_tabnam	CHAR(10)	Name of table structure	From IDoc control record or fixed "EDI_DC40"	X
sap_mandt	CHAR(3)	Client	From IDoc control record/ not relevant for processing in HYDRA	
sap_docnum	CHAR(16)	IDoc number	From IDoc control record/ not relevant for processing in HYDRA	
sap_docrel	CHAR(4)	SAP release for IDoc	From IDoc control record/ not relevant for processing in HYDRA	

Field name	Data type	Description	Purpose	Mandatory field
sap_status	CHAR(2)	Status of IDoc	From IDoc control record/ not relevant for processing in HYDRA	
sap_direct	CHAR(1)	Direction (point of view: R/3)	From IDoc control record/ not relevant for processing in HYDRA	
sap_outmod	CHAR(1)	Output mode of IDocs in R/3	From IDoc control record/ not relevant for processing in HYDRA	
sap_exprss	CHAR(1)	Overriding in inbound processing	From IDoc control record/ not relevant for processing in HYDRA	
sap_test	CHAR(1)	Test identifier	From IDoc control record/ not relevant for processing in HYDRA	
sap_idotyp	CHAR(30)	Name of basic type	According to specifications of the respective user data interface	X
sap_cimtyp	CHAR(30)	Extension (defined by customer) (sub-segment, e.g. for customizations --> future-proofed)	From IDoc control record/ not relevant for processing in HYDRA	
sap_mestyp	CHAR(30)	Message type	According to specifications of the respective user data interface	X
sap_mescod	CHAR(3)	Message code	From IDoc control record/ not relevant for processing in HYDRA	
sap_mesfct	CHAR(3)	Message function	According to specifications of the respective user data interface	X
sap_std	CHAR(1)	EDI standard, identifier	From IDoc control record/ not relevant for processing in HYDRA	
sap_stdvrs	CHAR(6)	EDI standard, version and release	From IDoc control record/ not relevant for processing in HYDRA	
sap_stdmes	CHAR(6)	EDI message type	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndpor	CHAR(10)	Sender port (SAP system, external subsystem)	From IDoc control record/ not relevant for processing in HYDRA	

Field name	Data type	Description	Purpose	Mandatory field
sap_sndprt	CHAR(2)	Partner type	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndpfc	CHAR(2)	Partner function of sender	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndprn	CHAR(10)	Partner number of sender (logical system)	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndsad	CHAR(21)	Sender address (SADR)	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndlad	CHAR(70)	Logical address of sender	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvpor	CHAR(10)	Receiver port	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvprt	CHAR(2)	Partner type	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvpfc	CHAR(2)	Partner function of recipient	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvprn	CHAR(10)	Partner number of receiver (log. system)	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvsad	CHAR(21)	Recipient address (SADR)	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvlad	CHAR(70)	Logical address of recipient	From IDoc control record/ not relevant for processing in HYDRA	
sap_credat	DATE	Created on (in ERP)	From IDoc control record/ not relevant for processing in HYDRA	
sap_cretim	INTEGER	Created at (in ERP)	From IDoc control record/ not relevant for processing in HYDRA	
sap_refint	CHAR(14)	Transmission file (EDI Interchange)	From IDoc control record/ not relevant for processing in HYDRA	

Field name	Data type	Description	Purpose	Mandatory field
sap_refgrp	CHAR(14)	Message group (EDI Message Group)	From IDoc control record/ not relevant for processing in HYDRA	
sap_refmes	CHAR(14)	Message (EDI Message)	From IDoc control record/ not relevant for processing in HYDRA	
sap_arckey	CHAR(70)	Key for external message archive	From IDoc control record/ not relevant for processing in HYDRA	
sap_serial	CHAR(20)	Serialization	From IDoc control record/ not relevant for processing in HYDRA	
param1	CHAR(40)	Additional parameters	Not used	
param2	CHAR(40)	Additional parameters	Not used	
bearb	CHAR(10)	Modified by	Not used	
bearb_date	DATE	Modified on	Not used	
bearb_time	INTEGER	Modified at	Not used	
verweis	Serial not null	Consecutive number	Assigned automatically by DB	

HYDRA inbound processing – table hysap_inbound_data

The data records of the IDoc (segments) are stored in the table "hysap_inbound_data". The transaction number represents the key for the tables "hysap_inbound_ctrl" and "hysap_inbound_data".

Field name	Data type	Description	Purpose	Mandatory field
ta_id	CHAR(30)	Transaction ID	Unique key Unique key (see above note)	X
ds_status	CHAR(3)	Segment status	Fixed "000"	X
ds_savdate	DATE	Date of receipt in HYERP	Current system date in format "mm/dd/yyyy"	X
ds_savtime	INTEGER	Time of receipt in HYERP	Current system time in seconds	X
ds_workdate	DATE	Date of the last editing	Not used / assigned	
ds_worktime	INTEGER	Time of the last editing	Not used / assigned	

Field name	Data type	Description	Purpose	Mandatory field
sap_segnam	CHAR(30)	Segment	According to specifications of the respective user data interface	X
sap_mandt	CHAR(3)	Client	From IDoc data record/ not relevant for processing in HYDRA	
sap_docnum	CHAR(16)	IDoc number	Reserved: fixed '0000000000000000'	X
sap_segnum	CHAR(6)	Segment number	Reserved: fixed '000000'	X
sap_psgnum	CHAR(6)	Number of the parent segment (if available)	Reserved; fixed: '000000'	X
sap_hlevel	CHAR(2)	Hierarchy level	Reserved; fixed: '00'	X
sap_sdata	CHAR(2000)	IDoc data	According to specifications of the respective user data interface	X
param1	CHAR(40)	Additional parameters	Not used	
param2	CHAR(40)	Additional parameters	Not used	
bearb	CHAR(10)	Modified by	Not used	
bearb_date	DATE	Modified on	Not used	
bearb_time	INTEGER	Modified at	Not used	
verweis	Serial not null	Consecutive number	Assigned automatically by DB	

HYDRA outbound processing – table hysap_out_ctrl

The table is only populated once data has been transferred successfully. The table is structured as follows.

Field name	Data type	Description	Example / comment	Mandatory field
ta_id	CHAR(30)	Transaction ID	Distinct transaction number (please see the above-mentioned note)	X
ta_type	CHAR(5)	Description of the structure type	fixed "IDOC"	X
ta_status	CHAR(3)	Processing status	fixed "099" (processed)	X

Field name	Data type	Description	Example / comment	Mandatory field
ta_lines	INTEGER	Number of segments of the data record included in the IDoc	Number of data records	X
ta_idone	INTEGER	Number of processed segments of the IDoc	Number of data records	X
sav_date	DATE	Date of receipt from HYDRA	Current system date in format "mm/dd/yyyy"	X
sav_time	INTEGER	Time of receipt from HYDRA	Current system time in "seconds after midnight"	X
work_dat	DATE	Date of the transfer	Current system date in format "mm/dd/yyyy"	X
work_time	TIME	Time of the transfer	Current system time in "seconds after midnight"	X
sap_tabnam	CHAR(10)	Name of table structure	Fixed "EDI_DC40"	X
sap_mandt	CHAR(3)	Client	From IDoc control record/ not relevant for processing in HYDRA	
sap_docnum	CHAR(16)	IDoc number	From IDoc control record/ not relevant for processing in HYDRA	
sap_docrel	CHAR(4)	SAP release for IDoc	From IDoc control record/ not relevant for processing in HYDRA	
sap_status	CHAR(2)	Status of IDoc	From IDoc control record/ not relevant for processing in HYDRA	
sap_direct	CHAR(1)	Direction (point of view: R/3)	From IDoc control record/ not relevant for processing in HYDRA	
sap_outmod	CHAR(1)	Output mode of IDocs in R/3	From IDoc control record/ not relevant for processing in HYDRA	
sap_exprss	CHAR(1)	Overriding in inbound processing	From IDoc control record/ not relevant for processing in HYDRA	
sap_test	CHAR(1)	Test identifier	From IDoc control record/ not relevant for processing in HYDRA	
sap_idoctyp	CHAR(30)	Name of basic type	According to specifications of the respective user data interface	X

Field name	Data type	Description	Example / comment	Mandatory field
sap_cimtyp	CHAR(30)	Extension (defined by customer) (sub-segment, e.g. for customizations --> future-proofed)	From IDoc control record/ not relevant for processing in HYDRA	
sap_mestyp	CHAR(30)	Message type	According to specifications of the respective user data interface	X
sap_mescod	CHAR(3)	Message code	From IDoc control record/ not relevant for processing in HYDRA	
sap_mesfct	CHAR(3)	Message function	According to specifications of the respective user data interface	X
sap_std	CHAR(1)	EDI standard, identifier	From IDoc control record/ not relevant for processing in HYDRA	
sap_stdvrs	CHAR(6)	EDI standard, version and release	From IDoc control record/ not relevant for processing in HYDRA	
sap_stdmes	CHAR(6)	EDI message type	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndpor	CHAR(10)	Sender port (SAP System, external subsystem)	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndprt	CHAR(2)	Partner type	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndpfc	CHAR(2)	Partner function of sender	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndprn	CHAR(10)	Partner number of sender (log. system)	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndsad	CHAR(21)	Sender address (SADR)	From IDoc control record/ not relevant for processing in HYDRA	
sap_sndlad	CHAR(70)	Logical address of sender	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvpor	CHAR(10)	Receiver port	From IDoc control record/ not relevant for processing in HYDRA	

Field name	Data type	Description	Example / comment	Mandatory field
sap_rcvprt	CHAR(2)	Partner type	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvpfc	CHAR(2)	Partner function of recipient	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvprn	CHAR(10)	Partner number of receiver (log. system)	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvsad	CHAR(21)	Recipient address (SADR)	From IDoc control record/ not relevant for processing in HYDRA	
sap_rcvlad	CHAR(70)	Logical address of recipient	From IDoc control record/ not relevant for processing in HYDRA	
sap_credat	DATE	Created on (in ERP)	From IDoc control record/ not relevant for processing in HYDRA	
sap_cretim	INTEGER	Created at (in ERP)	From IDoc control record/ not relevant for processing in HYDRA	
sap_refint	CHAR(14)	Transmission file (EDI Interchange)	From IDoc control record/ not relevant for processing in HYDRA	
sap_refgrp	CHAR(14)	Message group (EDI Message Group)	From IDoc control record/ not relevant for processing in HYDRA	
sap_refmes	CHAR(14)	Message (EDI Message)	From IDoc control record/ not relevant for processing in HYDRA	
sap_arckey	CHAR(70)	Key for external message archive	From IDoc control record/ not relevant for processing in HYDRA	
sap_serial	CHAR(20)	Serialization	From IDoc control record/ not relevant for processing in HYDRA	
param1	CHAR(30)	Additional parameters	Not used	
param2	CHAR(30)	Additional parameters	Not used	
bearb	CHAR(10)	Modified by	Not used	
bearb_date	DATE	Modified on	Not used	
bearb_time	INTEGER	Modified at	Not used	

Field name	Data type	Description	Example / comment	Mandatory field
verweis	Serial not null	Consecutive number	Assigned automatically by DB	

HYDRA outbound processing – table hysap_out_data

The table hysap_out_data is structured as follows.

Field name	Type	Description	Example / comment	Mandatory field
ta_id	CHAR(30)	Transaction ID	Distinct transaction number (please see the above-mentioned note)	X
ds_status	CHAR(3)	Segment status	Before the transfer "000" During the transfer "100" After a successful transfer, the status is changed to "099"	X
ds_savdate	DATE	Date of saving data	Assigned by HYDRA.	
ds_savtime	INTEGER	Time of saving data	Assigned by HYDRA.	
ds_workdate	DATE	Date of the transfer	Current system date in format "mm/dd/yyyy"	X
ds_worktime	INTEGER	Time of the transfer	Current system time in seconds	X
ds_source_sys	Char(10)	ERP target system	Assigned by HYDRA.	
sap_segnam	CHAR(30)	Segment	According to specifications of the respective user data interface / assigned by HYDRA	X
sap_mandt	CHAR(3)	Client	From IDoc data record/ not relevant for processing	
sap_docnum	CHAR(16)	IDoc number	From IDoc data record/ not relevant for processing	
sap_segnum	CHAR(6)	Segment number	From IDoc data record/ not relevant for processing	
sap_psgnum	CHAR(6)	Number of the parent segment (if available)	From IDoc data record/ not relevant for processing	
sap_hlevel	CHAR(2)	Hierarchy level	From IDoc data record/ not relevant for processing	

Field name	Type	Description	Example / comment	Mandatory field
sap_sdata	CHAR(1000)	IDoc data	According to specifications of the respective user data interface / assigned by HYDRA	X
param1	CHAR(40)	Additional parameters	Not used	
param2	CHAR(40)	Additional parameters	Not used	
bearb	CHAR(10)	Modified by	Not used	
bearb_date	DATE	Modified on	Not used	
bearb_time	INTEGER	Modified at	Not used	
verweis	Serial not null	Consecutive number	Assigned automatically by DB	
kopf_verweis	INTEGER	Header reference	Refers to the corresponding master segment in hierarchical structures	(X)

Archiving

Archiving of MLE inbound and outbound tables is based on MLE archiving.



Entries for inbound and outbound processing must be added to the distribution model in order to ensure archiving.